

1. Payoff Mechanics & Derivation

$$\text{Phi_net} = N * [(\text{FR_H} - \text{FR_L}) - 0.0020 \text{ (FeeDrag)} - 0.0010 \text{ (Slippage)} + \text{DeltaBasis}]$$

2. Delta-Neutrality Definition & Proof

$$N_L = Q_L \cdot P_L = N_H = Q_H \cdot P_H = \text{target_notional} / 2 \Rightarrow \text{Net Delta} = N_L - N_H = 0.0$$

3. Residual Risk Inventory

- Basis Risk: Divergence between venue A and venue B perp prices during 9.5-minute holding interval.
- Execution Timing Risk: Stochastic fill latency skew (40ms-350ms) causing momentary unhedged delta exposure.
- Funding Settlement Asynchrony: Timestamp discrepancies across venues or differing funding intervals.
- Exchange Outage / Disconnect: Single venue drops websocket feed leaving open leg unhedged.
- Auto-Deleveraging (ADL) / Liquidation Cascades: Extreme market moves triggering unhedged directional tail risk.

4. Sign-offs

- **GAMMA:** `GAMMA - 2026-08-28T14:52:28.651084+00:00 - git:m5-exec-2026`
- **BETA:** `BETA - 2026-08-28T14:52:28.651084+00:00 - git:m5-exec-2026`